

Coding & Solution

Date	28 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/akshitamanikchandra-dev/Rising-waters-AIML-Track
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy
Key Features Implemented	User Authentication, Flood Risk Prediction, ML Model Integration (Decision Tree, Random Forest, KNN, XGBoost), Prediction Dashboard, Prediction History, Database Storage, REST API
Pending / Incomplete Features	Cloud deployment optimization, Email/SMS alert notifications, Live weather API integration
Setup / Run Instructions	Clone the repository → Create a virtual environment → Install dependencies using pip install -r requirements.txt → Generate the ML model → Run python run.py → Open http://localhost:5000 in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate components for authentication, machine learning prediction, database management, and user interface. It uses multiple machine learning algorithms, with XGBoost achieving the highest prediction accuracy. The codebase is well-organized, reusable, and documented for future enhancements and deployment